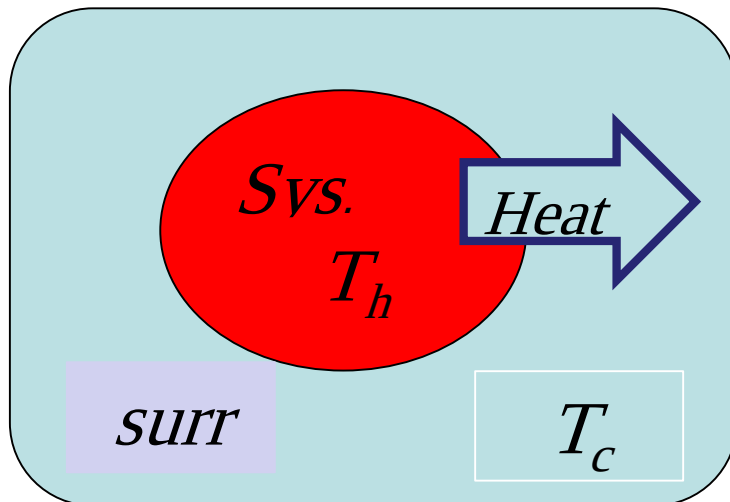


Chemical Thermodynamics

- The area of chemistry that deals with energy relationships, entropy, and the spontaneity of a chemical process.

19.1 Spontaneous Processes

- A spontaneous process is one that proceeds on its own without any outside assistance under given the set of conditions (T, P, conc. ,,,, et).



- Heat flows spontaneously from the hotter object to colder one but the reverse never happen spontaneously.

Gas at 1 atm

Evacuated flask 0 atm

Closed
stopcock

When stopcock opens,
gas expands to occupy
both flasks

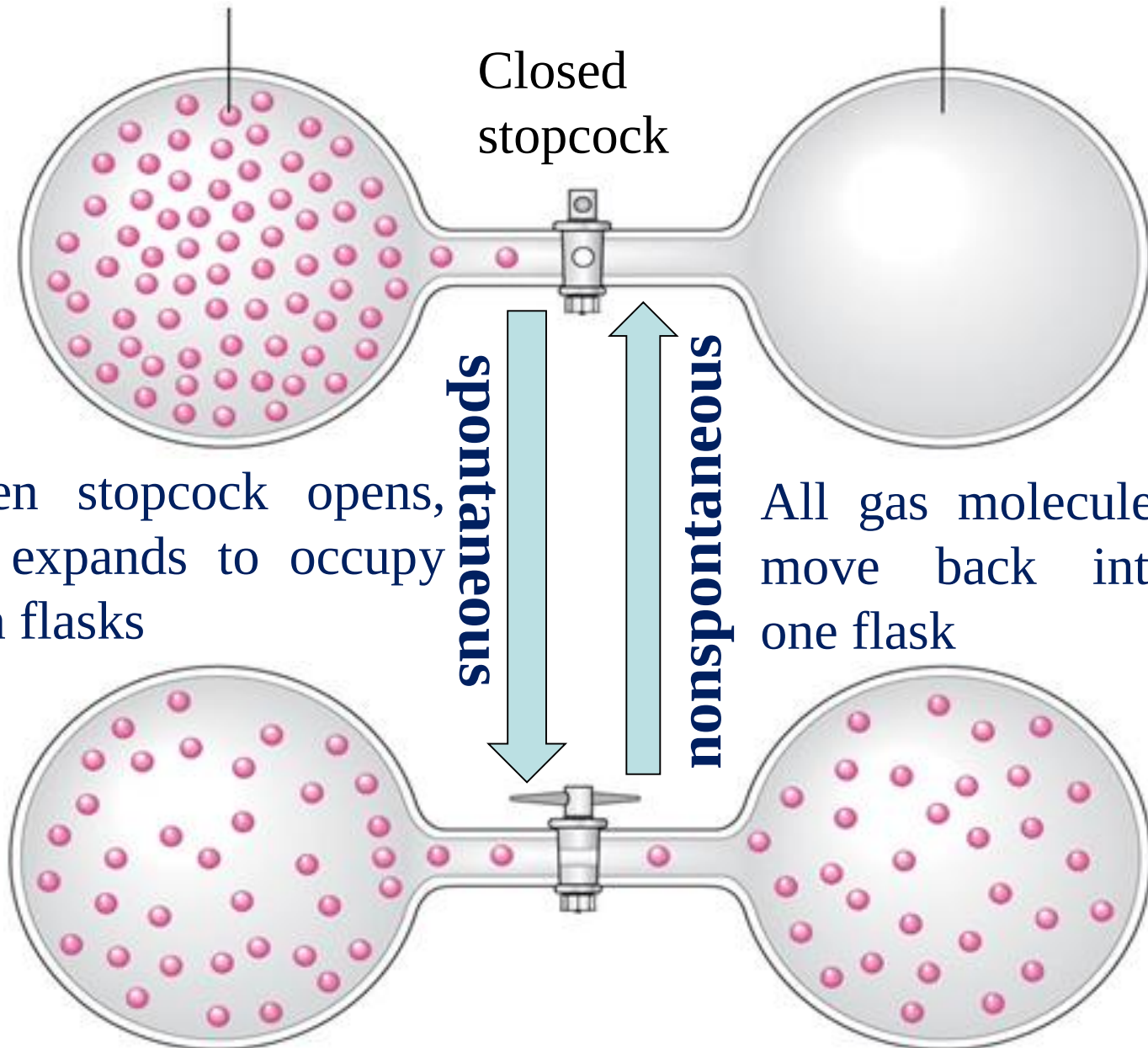
All gas molecules
move back into
one flask

spontaneous

nonspontaneous

0.5 atm

0.5 atm

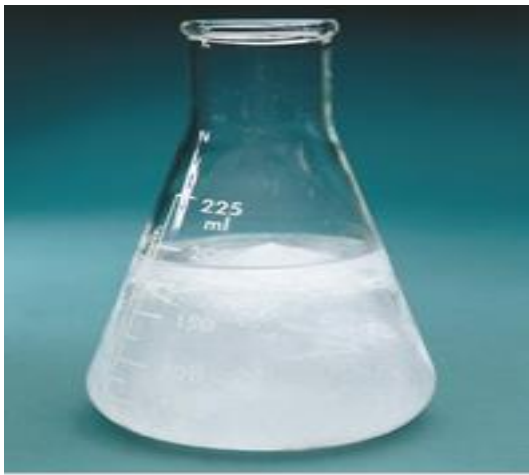




spontaneous
↓
↑
nonspontaneous



- In general, processes that are spontaneous in one direction are nonspontaneous in the opposite direction.
- A spontaneous reaction can be very fast, as in case of acid base neutralization, or very slow as in the rusting of iron.
- Thermodynamics tell us the direction and extent of a reaction but nothing about the speed.



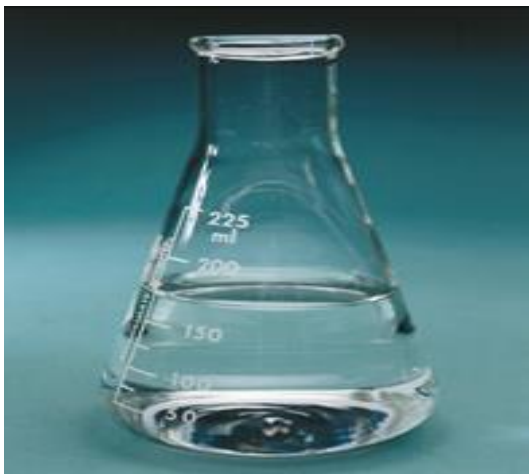
Spontaneous
at $T > 0.0$

↑

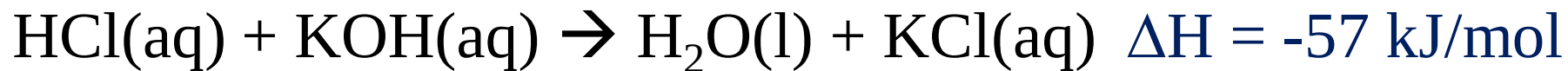
nonspontaneous

↑

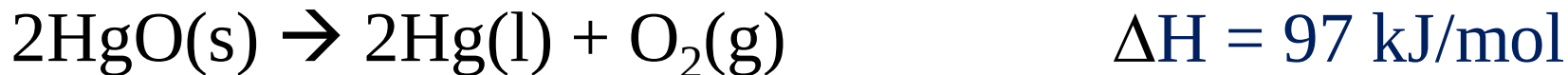
spontaneous
at $T < 0.0$



- Processes that are spontaneous at one temperature may be nonspontaneous at other temperatures.
- At 0°C , normal freezing point of water, the liquid and solid phases are in equilibrium.



❖ Often spontaneous processes are exothermic, but not always....



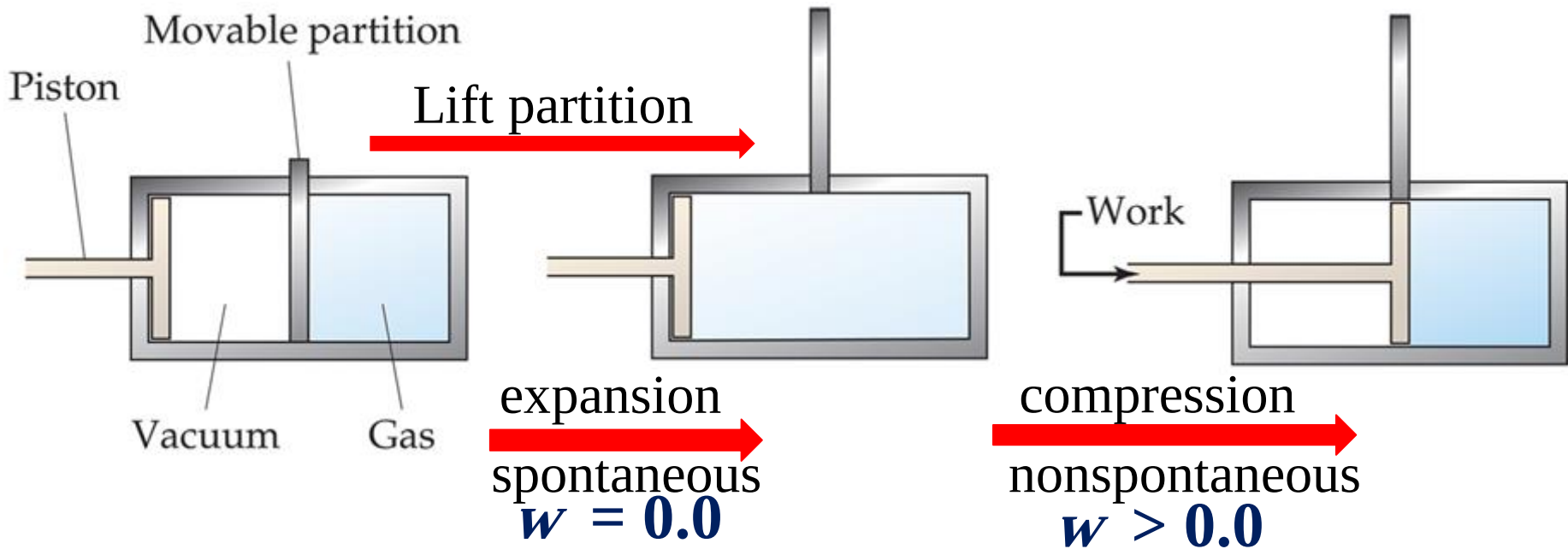
❖ Last two reactions occur spontaneously but these are an endothermic process...

❖ Exothermicity favors the spontaneity of a reaction but does not guarantee it.

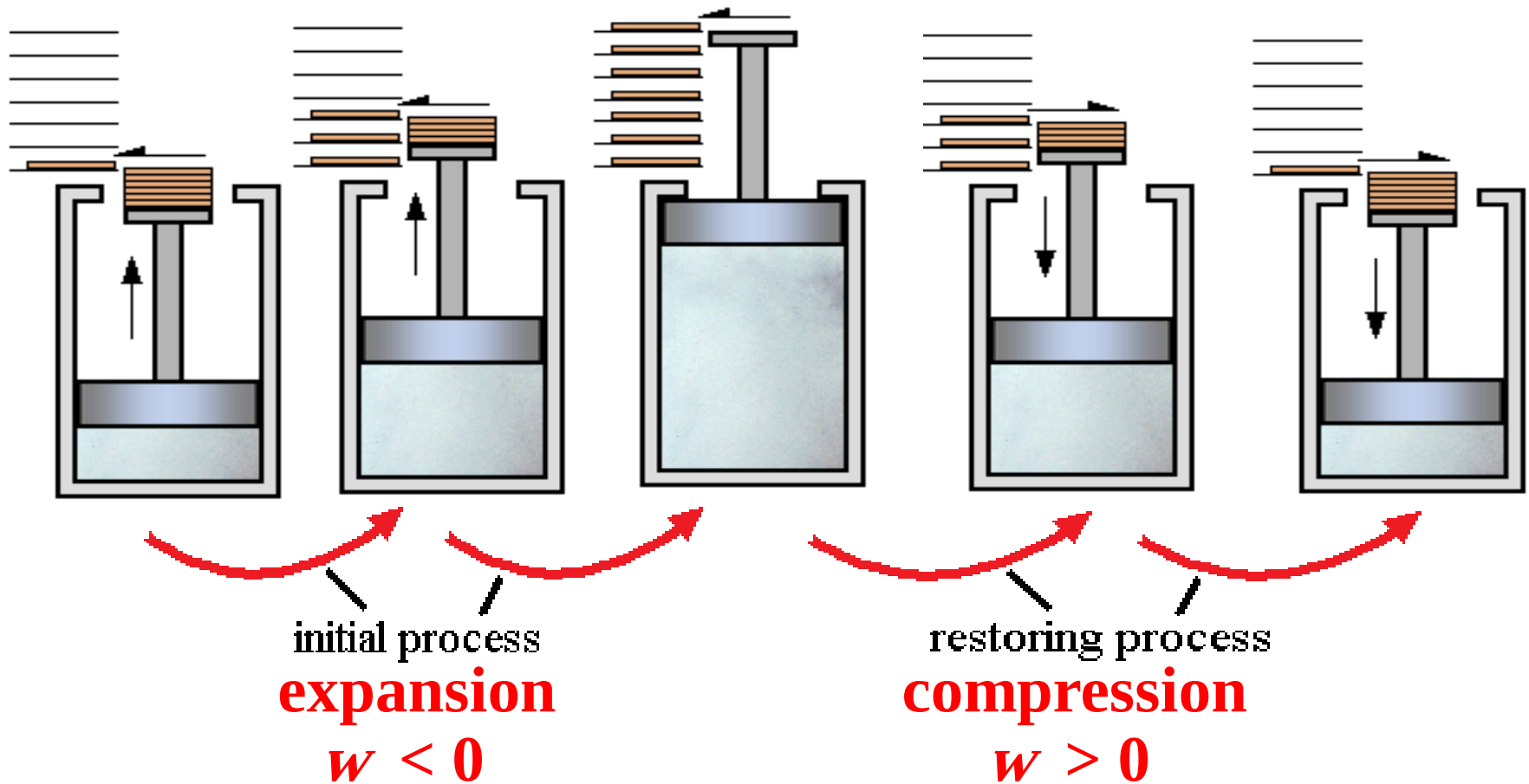
❖ There is another thermodynamic quantity!

Irreversible Processes

- An irreversible process is one that cannot simply be reversed to restore the system and its surroundings to original states.



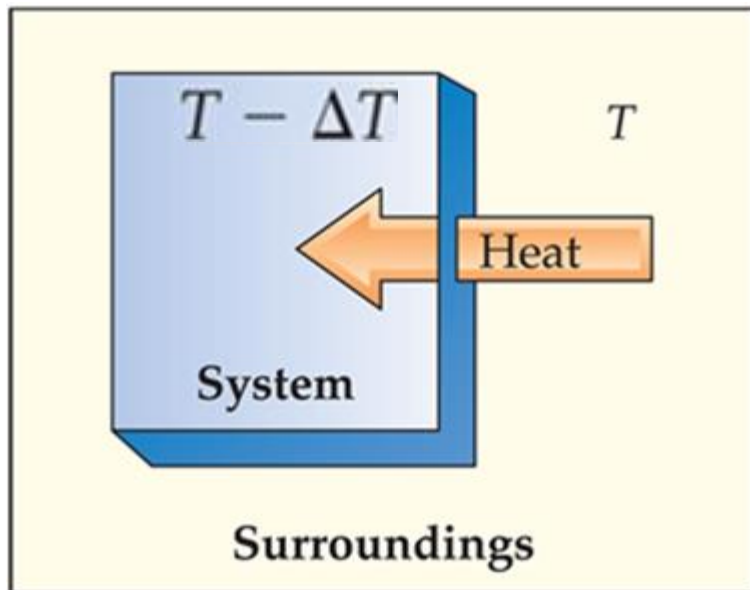
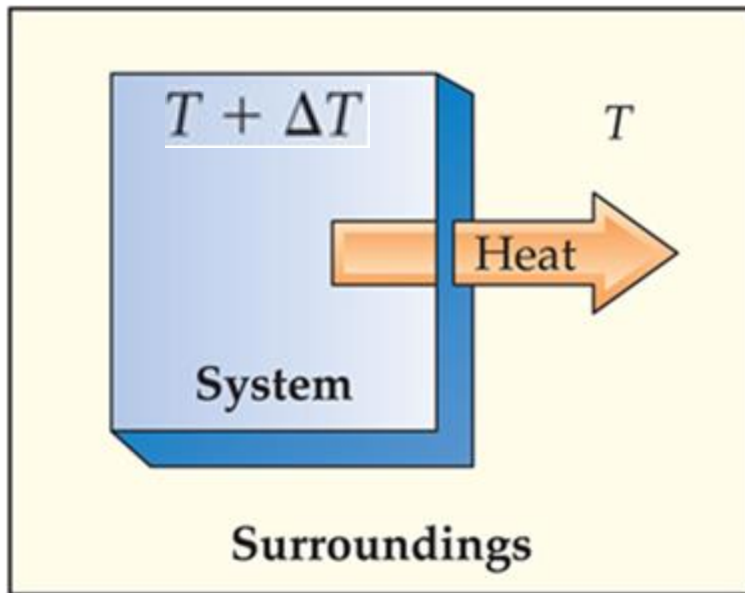
- All **Spontaneous** processes are **irreversible**.
- All **real** processes are **irreversible**.



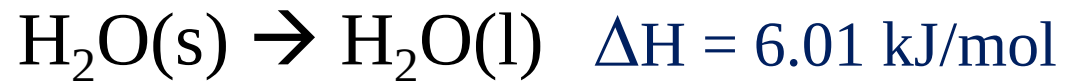
- In a reversible process the system changes in such a way that the system and surroundings can be restored to their original states by exactly reversing the change.

*Changes are infinitesimally **small** in a reversible process.*

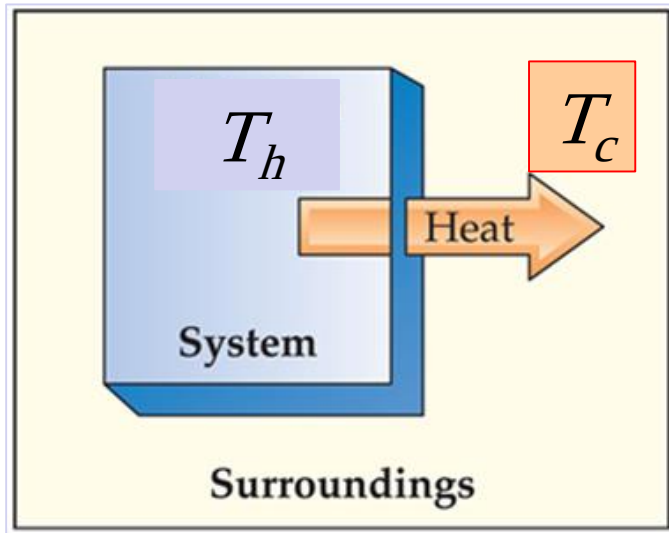
Reversible Processes



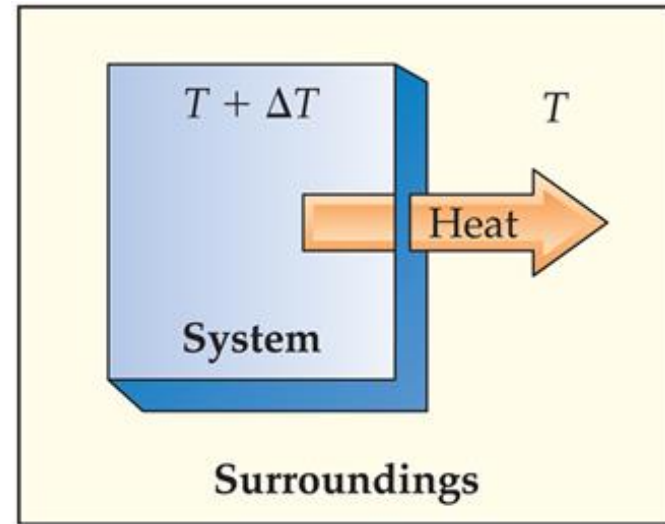
- Heat can flow between system and surrounding reversibly only if the two have infinitesimally small difference in temperature δT . This is obviously an idealization & can never happen in a real system!



Irreversible Process



Reversible Process



- Heat flows spontaneously from the hotter object to colder one in an irreversible process.
- Heat can flow between system and surrounding reversibly only if the two have infinitesimally small difference in temperature δT .

Reversible processes are those that reverse direction whenever an infinitesimal change is made in some property of the system.



19.2 Entropy

- **Entropy (S):** A thermodynamic function that describes the degree of randomness or probability of existence.
- The entropy, S , of a system is a state function, therefore, $\Delta S = S_{\text{final}} - S_{\text{initial}}$.
- For a process occurring at constant temperature (an isothermal process):

$\text{J/K} \leftarrow \Delta S = \frac{q_{\text{rev}}}{T}$

heat that would be transfer if the process were reversible (J)

the absolute temperature, K

- Because S is a state function, we can use this equation to calculate ΔS for any isothermal process between states, not just the reversible one.

ΔS for phase changes

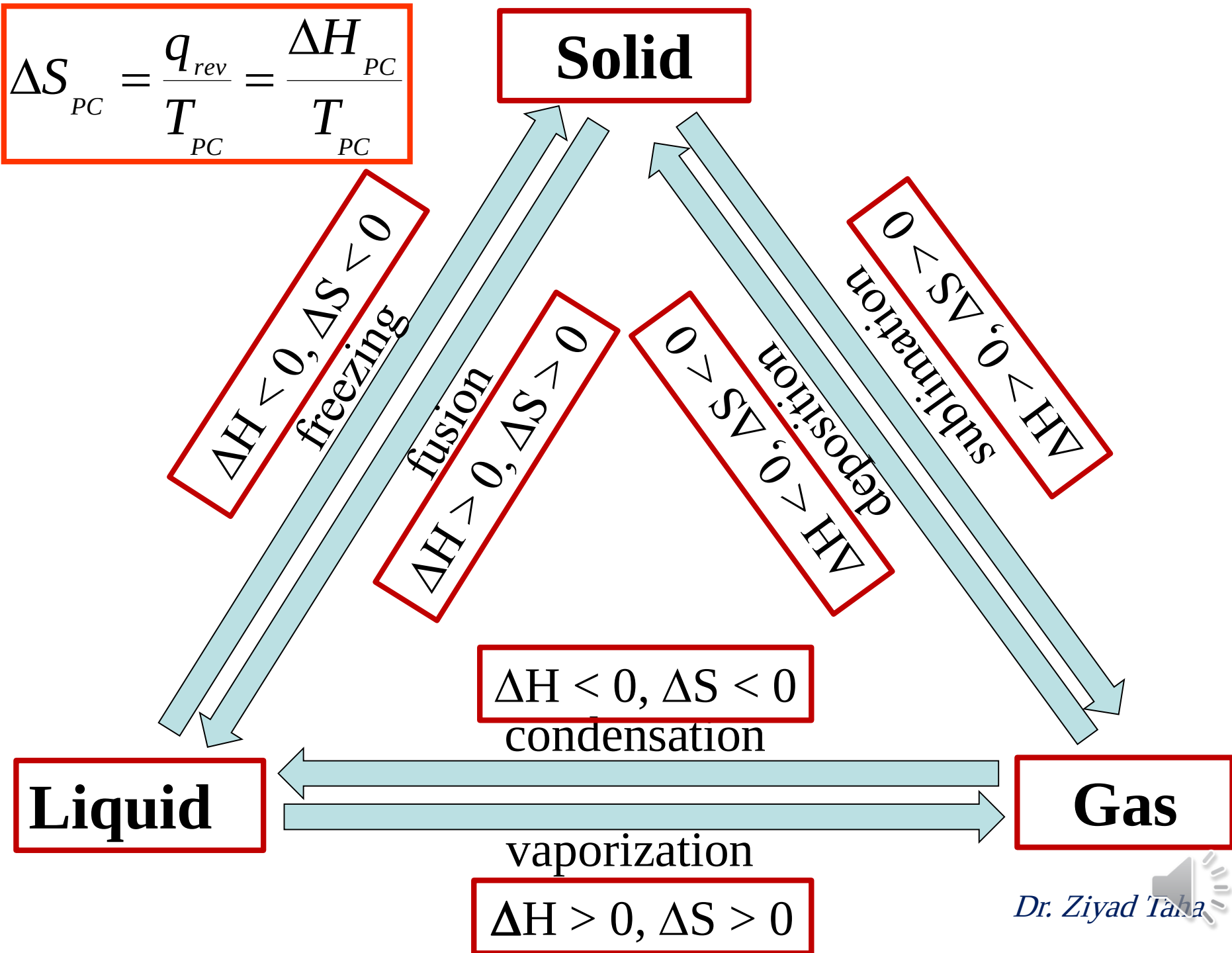
- Melting of a substance at its melting point and the vaporization of a substance at its boiling point are reversible isothermal processes.

$$\Delta S_{\text{fusion}} = \frac{q_{\text{rev}}}{T_m} = \frac{\Delta H_{\text{fusion}}}{T_m} \longrightarrow \text{melting point}$$

$$\Delta S_{\text{vap}} = \frac{q_{\text{rev}}}{T_b} = \frac{\Delta H_{\text{vap}}}{T_b} \longrightarrow \text{boiling point}$$

$$\Delta S_{\text{PC}} = \frac{q_{\text{rev}}}{T_{\text{PC}}} = \frac{\Delta H_{\text{PC}}}{T_{\text{PC}}} \longrightarrow \text{phase change temperature}$$

$$\Delta S_{PC} = \frac{q_{rev}}{T_{PC}} = \frac{\Delta H_{PC}}{T_{PC}}$$



Sample Exercises

1. Calculate the molar entropy of vaporization of H_2O ? The molar enthalpy of vaporization for water is 44 kJ/mol and the boiling point is 100°C .

$$\Delta S_{\text{vap}} = \frac{q_{\text{rev}}}{T_b} = \frac{\Delta H_{\text{vap}}}{T_b} \longrightarrow \Delta S_{\text{vap}} = \frac{44 \times 10^3 \text{ J}}{(100 + 273) \text{ K}}$$
$$= 118 \text{ J/mol.K}$$

2. Calculate the molar entropy of fusion of H_2O ? The molar enthalpy of fusion for water is 6.01 kJ/mol and the freezing point is 0.0°C .

$$\Delta S_{\text{fusion}} = \frac{q_{\text{rev}}}{T_m} = \frac{\Delta H_{\text{fusion}}}{T_m} \longrightarrow \Delta S_{\text{fusion}} = \frac{6.01 \times 10^3 \text{ J}}{(0 + 273) \text{ K}}$$
$$= 22.0 \text{ J/mol K}$$

Sample Exercise: The normal boiling point of $\text{C}_2\text{H}_5\text{OH}(\text{g})$ (M. mass = 46 g/mol) is 78.3°C , and its $\Delta H_{\text{vap}} = 38.56 \text{ kJ/mol}$. What is the change in entropy of the system when 68.3 g of $\text{C}_2\text{H}_5\text{OH}(\text{g})$ at 1.0 atm condenses to liquid at the 78.3°C .

$$\Delta S_{\text{vap}} = \frac{q_{\text{rev}}}{T_b} = \frac{\Delta H_{\text{vap}}}{T_b}$$

$$\Delta S_{\text{vap}} = \frac{38.56 * 10^3 \text{ J/mol}}{(78.3 + 273)\text{K}}$$

$$= -109.8 \text{ J/mol.K}$$

$$\Delta S_{\text{vap}} = -109.8 \frac{\text{J}}{\text{mol.k}} \times \frac{\text{mol C}_2\text{H}_5\text{OH}}{46 \text{ g}} \times 68.3 \text{ g}$$

$$= -163 \text{ J/mol.K}$$

Sample Exercise

If the molar enthalpy of fusion for water is 6.01 kJ/mol and the freezing point is 0.0°C. Calculate the entropy change for the system and its surroundings

A. when 1 mol of ice melting is placed at the body temperature 37°C?

$$\Delta S_{\text{sys}} = 22.0 \text{ J/mol. K.}$$

$$\Delta S_{\text{surr}} = -19.4 \text{ J/mol. K.}$$

$$\Delta S_{\text{univ}} = 2.6 \text{ J/mol. K.}$$

→ spontaneous

B. when 1 mol of ice is placed at 0°C?

$$\Delta S_{\text{sys}} = 22.0 \text{ J/mol. K.}$$

$$\Delta S_{\text{surr}} = -22.0 \text{ J/mol. K.}$$

$$\Delta S_{\text{univ}} = 0.0 \text{ J/mol. K.}$$

→ reversible process

Second Law of Thermodynamics

The entropy of the universe does not change for reversible processes and increases for spontaneous processes.

- *Reversible process:*

$$\Delta S_{univ} = \Delta S_{sys} + \Delta S_{surr} = 0$$

- *Irreversible (real, spontaneous):*

$$\Delta S_{univ} = \Delta S_{sys} + \Delta S_{surr} > 0$$

- *Irreversible (nonspontaneous):*

$$\Delta S_{univ} = \Delta S_{sys} + \Delta S_{surr} < 0$$

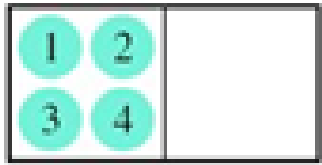


19.3 Molecular Interpretation of Entropy

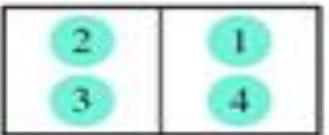
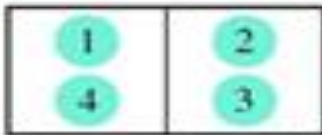
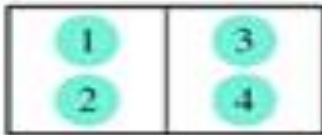
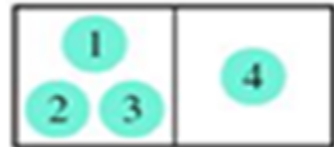
Boltzmann Equation and Microstates

Microstates

Distribution



I



II

III

- Distribution III is most probable because there are 6 microstates or 6 ways to achieve it.
- Distribution I is least probable because it has one microstate and there is only one way to achieve it.

- Entropy (S) is

$$S = k \ln W$$

- k : Boltzmann constant, 1.38×10^{-23} J/K.
- W : # of microstates

- The larger the W , the greater the entropy.

$$\Delta S = S_f - S_i$$

$$\Delta S = k \ln W_f - k \ln W_i$$

$$\Delta S = k \ln \frac{W_f}{W_i}$$

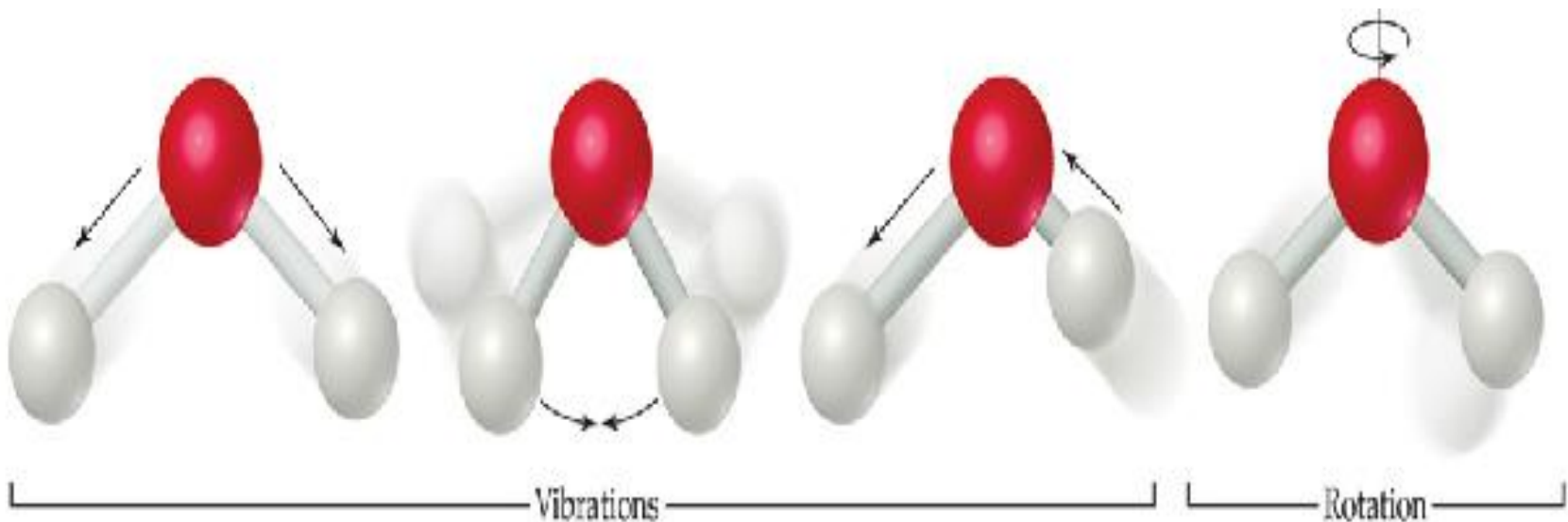
- $W_f > W_i$ then $\Delta S > 0$ entropy of the system increases

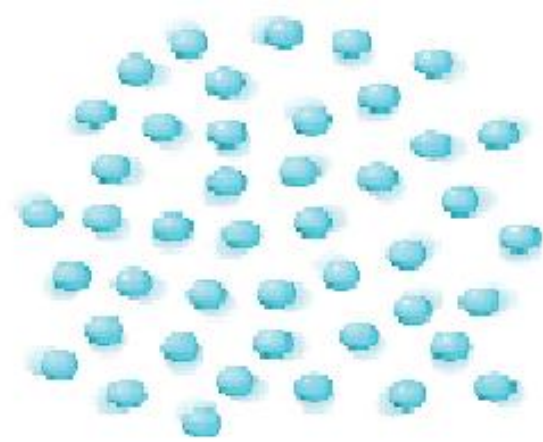
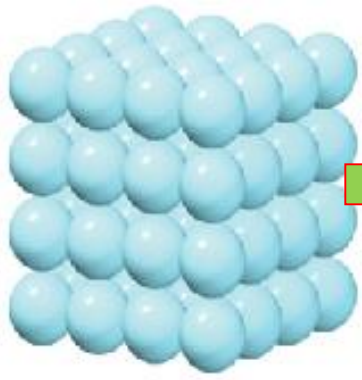


- The entropy for a system increases with:
 - an increase in the number of molecules.
 - an increase in volume.
 - an increase in temperature.

- Because any of these changes increases the number of the possible positions available and kinetic energies of the molecules making the system which increases the number of microstates possible (entropy) for the system.

- Molecules exhibit several types of motion:
 - Translational: movement of the entire molecule from one place to another.
 - Vibrational: periodic motion of atoms within a molecule.
 - Rotational: rotation of the molecule on about an axis or rotation about σ bonds.





$H_2O(s)$

motion restricted
to vibration only

smallest # of
microstate

$H_2O(l)$

increased freedom
with respect to
translation. Freed
to vibrate & rotate.

larger # of
microstate

$H_2O(g)$

complete freedom
for translation,
vibration and
rotation

largest # of
microstate

$$S_{\text{solid}} < S_{\text{liquid}} < S_{\text{gas}}$$





NaCl(s)

solution process
H₂O

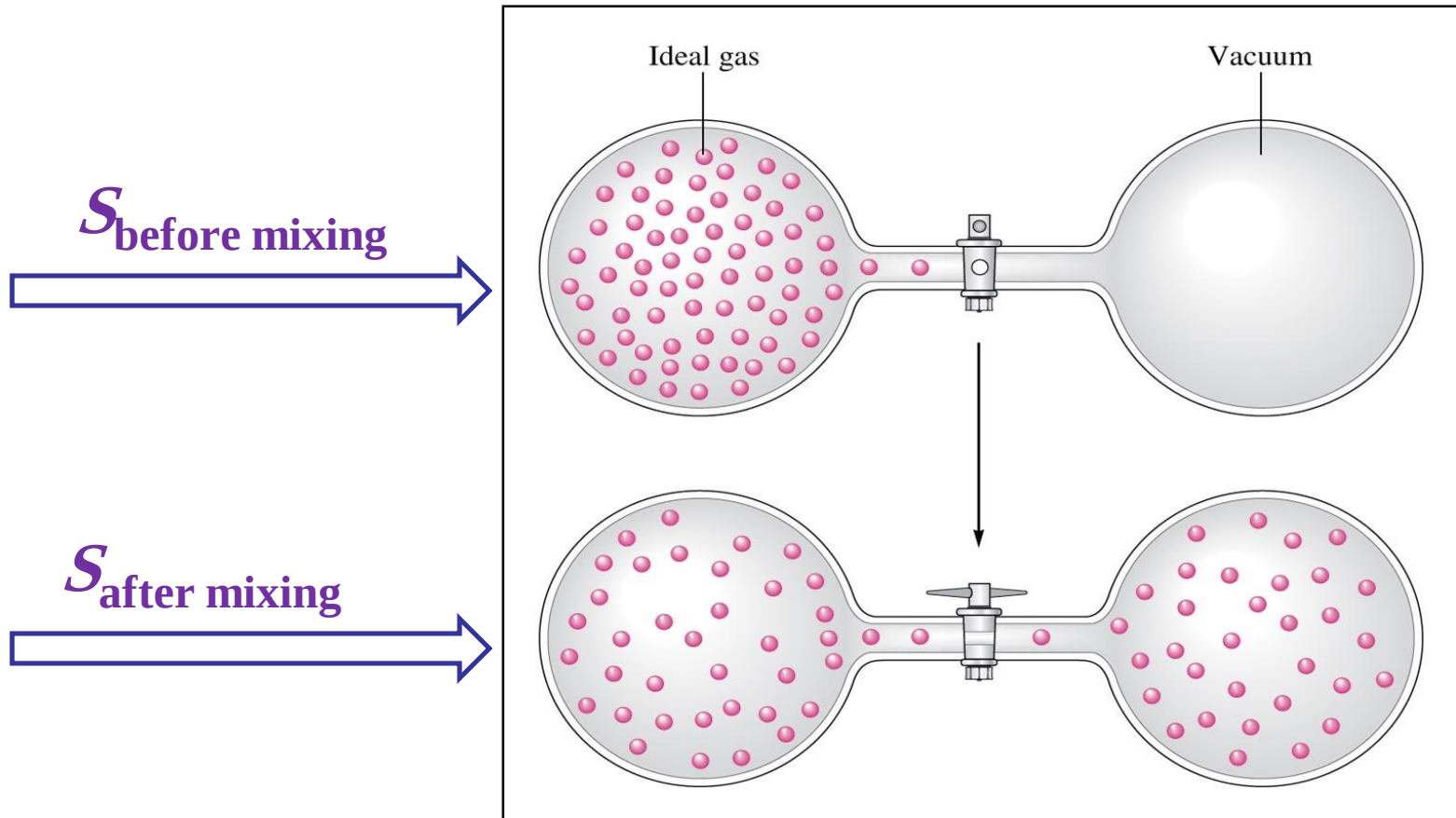
$$S_{\text{NaCl(s)}} < S_{\text{NaCl(aq)}}$$



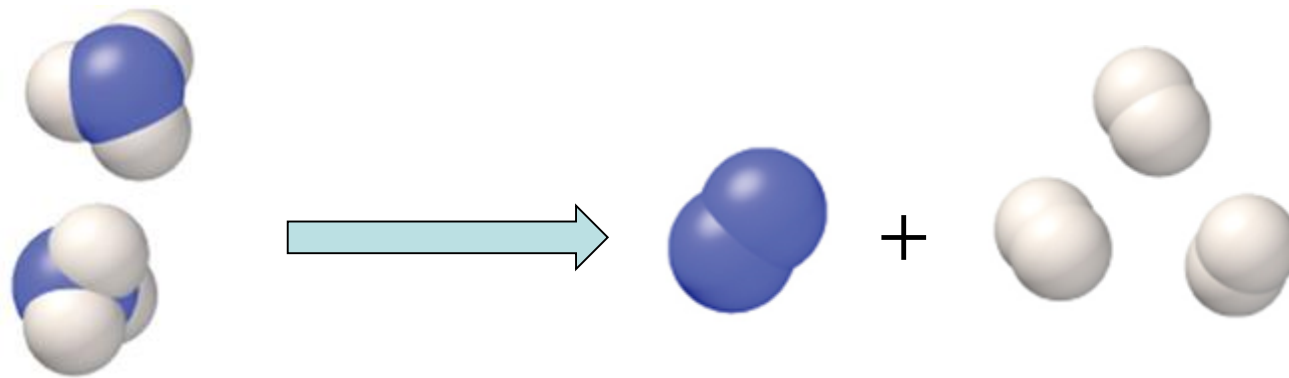
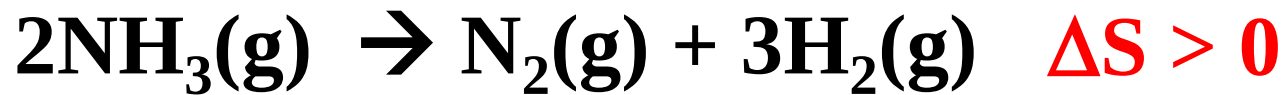
NaCl(aq)

- **Solution process involves both:**
 - a) disordering process: the ions become less confined.
 - b) ordering process: some H₂O molecules become more confined.
- The disordering process are usually dominant, and so the overall effect is an increase in the randomness of the system.

The Expansion of an ideal gas into an evacuated bulb



$$S_{\text{after mixing}} > S_{\text{before mixing}} \Rightarrow \Delta S > 0.0$$



2 molecules

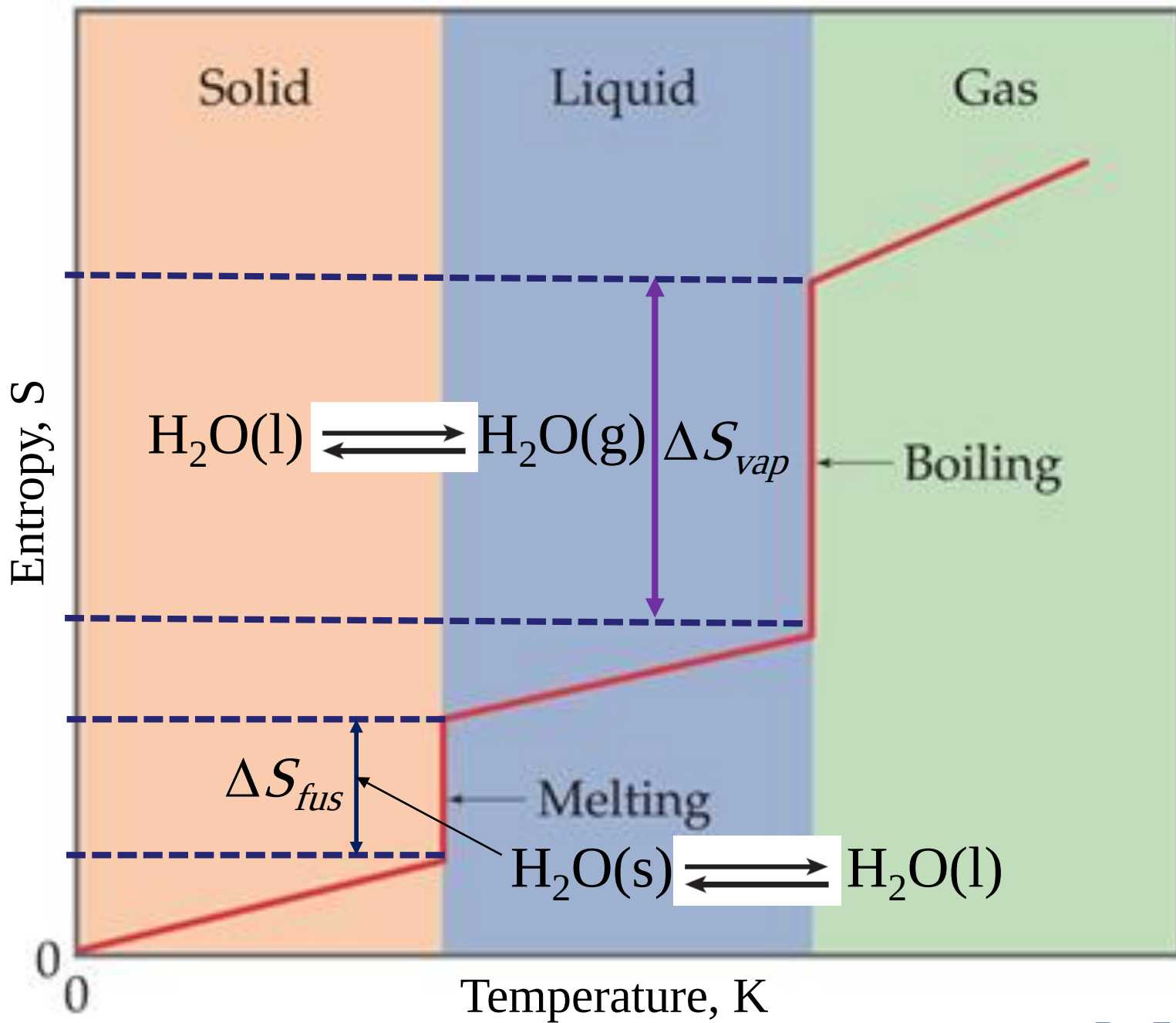


4 molecules

$$\Delta n_g = \sum n_{g_{\text{product}}} - \sum n_{g_{\text{reactants}}}$$

- If $\Delta n_g > 0 \rightarrow \Delta S > 0$.
- If $\Delta n_g < 0 \rightarrow \Delta S < 0$.





Third Law of Thermodynamics

The entropy of a pure perfect crystalline substance is zero at absolute zero.

$$S(0 \text{ K}) = 0$$

- At 0 K, the individual atoms or molecules in the lattice would be perfectly ordered and as well defined in position and so $W = 1$.

$$S_{0K} = k \ln 1 = 0 \longrightarrow S_{0K} = k \ln 1 = 0$$

- Importance of this law: it allows us to calculate absolute entropies for substances.

$$\Delta S = S_f - S_i \longrightarrow \Delta S = S_f - S_{0K}$$

$$\Delta S = S_f - 0 \longrightarrow \Delta S = S_f$$

19.4 Entropy Changes in Chemical Reactions

- **Standard molar entropies, S° :** molar entropies for substance in their standard states.
- S° of elements at 298 K > zero.
- $S^\circ(\text{g}) > S^\circ(\text{l}) > S^\circ(\text{s})$.
- S° generally increase with increasing molar mass.
- S° generally increase with an increasing number of atoms in the formula of a substance.

Substance	S , J/mol.K
Li(s)	29.1
Na(s)	51.4
H ₂ (g)	130.6
H₂O(g)	188.8
H₂O(l)	69.9
CH ₃ OH(l)	126.8
C ₆ H ₆ (l)	172.8
CH ₄ (g)	186.3
C ₂ H ₆ (g)	229.6
C ₃ H ₈ (g)	270.3

- Standard entropy change in a chemical reaction, $\Delta S^\circ_{\text{rxn}}$

$$\Delta S^\circ_{\text{rxn}} = \sum n S^\circ_{\text{products}} - \sum m S^\circ_{\text{reactants}}$$

○ n, m: are the coefficient in the balanced chemical equation.

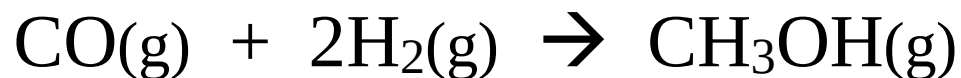
Sample exercise:



S° , J/K mol: 270 205 214 189

$$\begin{aligned} \Delta S^\circ_{\text{rxn}} &= \{3 \times S^\circ_{\text{CO}_2} + 4 \times S^\circ_{\text{H}_2\text{O}}\} - \{S^\circ_{\text{C}_3\text{H}_8} + 5 \times S^\circ_{\text{O}_2}\} \\ &= \{3 \times 214 + 4 \times 189\} - \{1 \times 270 + 5 \times 205\} \\ &= 103 \text{ J/K} \quad > 0, \Delta n_g = 1 \end{aligned}$$

Sample exercise:



$$S^\circ, \text{ J/K mol: } \quad 198 \quad \quad 131 \quad \quad 240$$

$$\begin{aligned}\Delta S^\circ_{\text{rxn}} &= \{1 \times S^\circ_{\text{CH}_3\text{OH}}\} - \{1 \times S^\circ_{\text{CO}} + 2 \times S^\circ_{\text{H}_2}\} \\ &= \{1 \times 240\} - \{1 \times 198 + 2 \times 131\} \\ &= -220 \text{ J/K} \quad \Delta S^\circ_{\text{rxn}} < 0 \text{ if } \Delta n_g = -2\end{aligned}$$

Sample exercise:



$$S^\circ, \text{ J/K mol: } \quad 92.9 \quad \quad 39.8 \quad \quad 213.6$$

$$\begin{aligned}\Delta S^\circ_{\text{rxn}} &= \{1 \times S^\circ_{\text{CO}_2} + 1 \times S^\circ_{\text{CaO}}\} - \{1 \times S^\circ_{\text{CaCO}_3}\} \\ &= \{1 \times 213.6 + 1 \times 39.8\} - \{1 \times 92.9\} \\ &= 160.5 \text{ J/K. mol} \quad \Delta S^\circ_{\text{rxn}} > 0 \text{ if } \Delta n_g = 1\end{aligned}$$

Entropy Changes in the Surroundings

- Tabulated absolute entropy values can be used to calculate the entropy change in a system (in a chemical rxn). **What about the entropy in the surroundings??**

$$\Delta S_{surr} = \frac{-q_{sys}}{T_{surr}}$$

Sample exercise:

Given the following information's:



S° , J/K mol: 191.5 130.6 192.5

Calculate $\Delta S^\circ_{\text{univ}}$.

$$\Delta S^\circ_{\text{rxn}} = \{2 \times S^\circ_{\text{NH}_3}\} - \{1 \times S^\circ_{\text{N}_2} + 3 \times S^\circ_{\text{H}_2}\}$$

$$\Delta S^\circ_{\text{rxn}} = \{2 \times 192.5\} - \{1 \times 191.5 + 3 \times 130.6\}$$

$\Delta S^\circ_{\text{rxn}} =$	-198 J/K	$= \Delta S^\circ_{\text{sys}}$
---------------------------------	--------------------	---------------------------------



$$\Delta S_{surr} = \frac{-q_{sys}}{T_{surr}}$$

$$\Delta S_{surr} = \frac{-(-92.4 * 1000 \text{ J})}{(25 + 273) \text{ K}}$$

$$\Delta S^{\circ}_{surr} = 310 \text{ J/mol. K}$$

$$\Delta S^{\circ}_{univ} = \Delta S^{\circ}_{sys} + \Delta S^{\circ}_{surr}$$

$$\Delta S^{\circ}_{univ} = -198 + 310$$

$$\Delta S^{\circ}_{univ} = 112 \text{ J/K} > 0.0 \rightarrow \text{spontaneous.}$$

19.5 Gibbs Free Energy (G)

- 2nd Law of thermodynamic for spontaneous process:

$$\Delta S_{univ} = \Delta S_{sys} + \Delta S_{surr} > 0$$

$$\Delta S_{univ} = \Delta S_{sys} + \frac{-\Delta H_{sys}}{T} > 0$$

$$-T\Delta S_{univ} = -T\Delta S_{sys} + \Delta H_{sys} < 0$$

$$-T\Delta S_{univ} = \Delta H_{sys} - T\Delta S_{sys} < 0$$

$$G = H - TS$$

- Since H , T , & S are state of functions, G is a state of function

$$\Delta G = \Delta H - T\Delta S$$

$$-T\Delta S_{univ} = \Delta G_{sys} = -T\Delta S_{sys} + \Delta H_{sys} < 0$$

- Under standard conditions: $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$



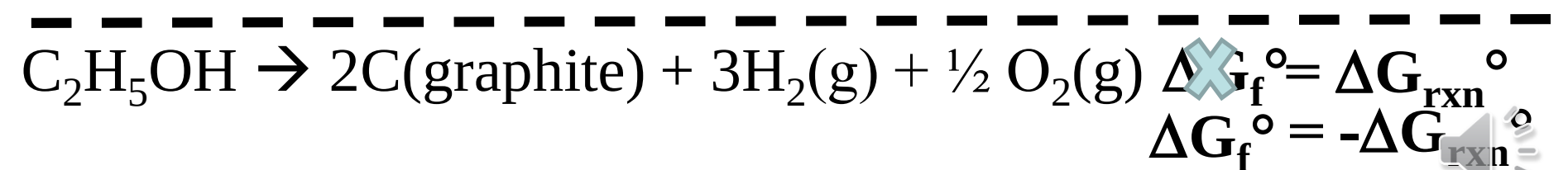
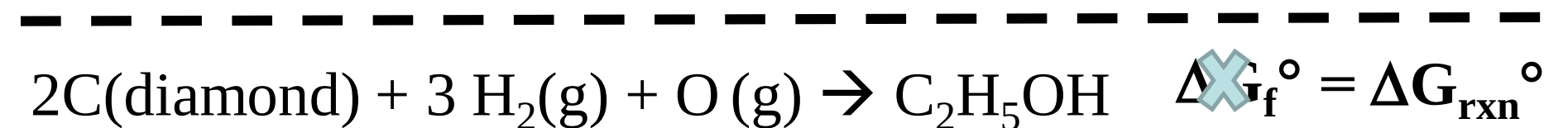
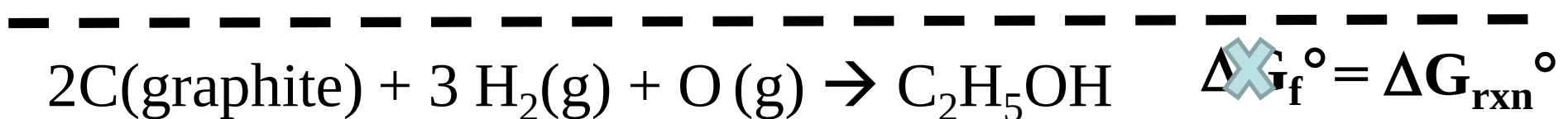
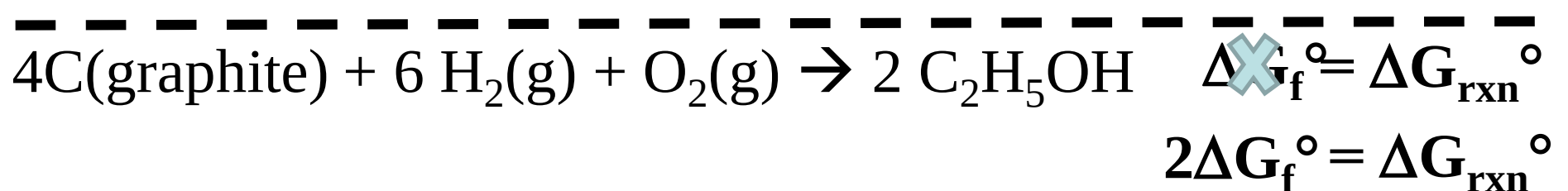
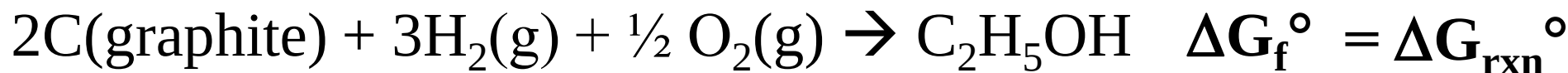
Gibbs Free Energy

- If $\Delta G < 0$, negative  *the forward reaction is spontaneous*
- If $\Delta G > 0$, positive  *the forward reaction is nonspontaneous*
- If $\Delta G = 0$  *the reaction is at equilibrium*

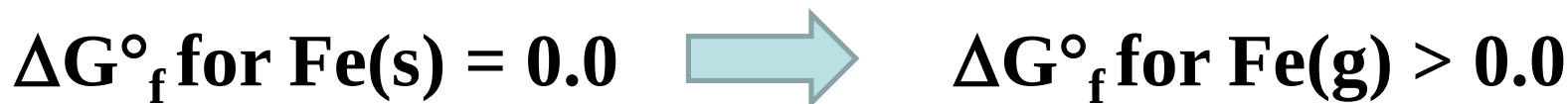
■ *In any spontaneous process carried out at constant T and P , the free energy always decreases.*

Standard Free Energy of Formation

- ΔG_f° Standard free energies of formation of a **compound** is defined as the free energy change for the reaction that forms **one mole** of the compound from its **elements in their most stable form** at standard states.



- ΔG_f° of **elements** in their most stable form at standard condition is **zero**.



- **Standard free energy for chemical processes:**

$$\Delta G^\circ_{rxn} = \sum n \cdot \Delta G_f^\circ(\text{products}) - \sum n \cdot \Delta G_f^\circ(\text{reactants})$$

- **note: n & m are stoichiometric coefficients.**

Sample Exercise

Calculate the standard free-energy change for the following reaction

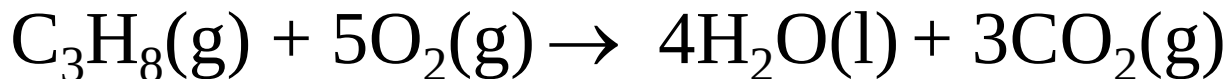


$$\begin{array}{ccc} \Delta G^\circ_f, \text{ kJ/mol} & -289.9 & -408.3 \\ \hline \end{array}$$

$$\begin{aligned} \Delta G^\circ_{\text{rxn}} &= \sum n \Delta G^\circ_{\text{products}} - \sum m \Delta G^\circ_{\text{reactants}} \\ &= \{2 \times -403 + 3 \times 0\} - \{2 \times -289.9\} \\ &= -236.8 \text{ kJ/mol (spont)} \end{aligned}$$

Sample Exercise

Calculate the standard free-energy change for the following reaction



$$\Delta G^\circ_f, \text{ kJ/mol} \quad \begin{array}{cccc} & -23.47 & & -237.14 & -394.4 \\ \hline \end{array}$$

$$\Delta G^\circ_{\text{rxn}} = \sum n \Delta G^\circ_{\text{products}} - \sum m \Delta G^\circ_{\text{reactants}}$$

$$= \{3 \times -394.4 + 4 \times -237.14\} - \{1 \times -23.47 + 5 \times 0\}$$

$$= -2108 \text{ kJ/mol (spont)}$$

19.6 Free Energy and Temperature

- Predict whether ΔG° for the following reaction is negative or positive?



The reaction is exothermic $\Delta H(-)$
 $\Delta n_g = 3 - 2 = 1 > 0 \rightarrow \Delta S (+)$



$$\begin{aligned}\Delta G &= \Delta H - T\Delta S \\ &= (-) - T(+)\end{aligned}$$

$= (-)$ at all T

- Predict whether ΔG° for the following reaction is negative or positive?



The reaction is endothermic $\Delta H(+)$
 $\Delta n_g = 2 - 3 = -1 < 0 \rightarrow \Delta S (-)$



$$\begin{aligned}\Delta G &= \Delta H - T\Delta S \\ &= (+) - T(-)\end{aligned}$$

$= (+)$ at all T

- Predict whether ΔG° for the following reaction is negative or positive?

$$\text{H}_2\text{O(l)} \rightarrow \text{H}_2\text{O(s)}$$

The reaction is exothermic $\Delta H(-)$

$$S_{\text{H}_2\text{O(l)}} > S_{\text{H}_2\text{O(s)}} \rightarrow \Delta S (-)$$

$$\begin{aligned} \Delta G &= \Delta H - T\Delta S && \Rightarrow = (-) \text{ at low } T && \Rightarrow T < 0^\circ\text{C} \\ &= (-) - T(-) && \Rightarrow = (+) \text{ at high } T && \Rightarrow T > 0^\circ\text{C} \end{aligned}$$

- Predict whether ΔG° for the following reaction is negative or positive?

$$\text{H}_2\text{O(s)} \rightarrow \text{H}_2\text{O(l)}$$

The reaction is endothermic $\Delta H(+)$

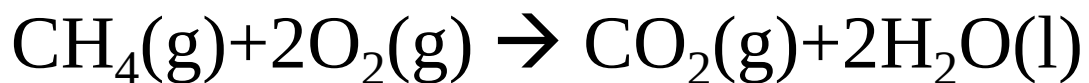
$$S_{\text{H}_2\text{O(l)}} > S_{\text{H}_2\text{O(s)}} \rightarrow \Delta S (+)$$

$$\begin{aligned} \Delta G &= \Delta H - T\Delta S && \Rightarrow = (-) \text{ at high } T && \Rightarrow T > 0^\circ\text{C} \\ &= (+) - T(+) && \Rightarrow = (+) \text{ at low } T && \Rightarrow T < 0^\circ\text{C} \end{aligned}$$

How signs of ΔH & ΔS affect rxn spontaneity

ΔH	ΔS	$-T\Delta S$	ΔG	Reaction Characteristic	Example
-	+	-	-	Spon. at all T	$2 \text{O}_3(\text{g}) \rightarrow 3 \text{O}_2(\text{g})$
+	-	+	+	Nonspon. at all T	$3 \text{O}_2(\text{g}) \rightarrow 2 \text{O}_3(\text{g})$
-	-	+	-	Spon. at low T	$\text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_2\text{O}(\text{s})$
			+	Nonspon. at high T	
+	+	-	-	Spon. at high T	$\text{H}_2\text{O}(\text{s}) \rightarrow \text{H}_2\text{O}(\text{l})$
			+	Nonspon. at low T	

- Predict whether $\Delta^\circ G$ for the following reaction is negative or positive?



$\Delta H (-)$



= (+) at high T

$\Delta S (-)$



= (-) at low T



$\Delta H (+)$



= (-) at high T

$\Delta S (+)$



= (+) at low T

Sample Exercise

For a reaction in which $\Delta H = 125 \text{ kJ/mol}$ and $\Delta S = 325 \text{ J/K}\cdot\text{mol}$ at 298 K.

a) Is the reaction spontaneous at 25°C.

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$\Delta G^\circ = 125000 \text{ J/mol} - 298 \text{ K} \times 325 \text{ J/mol}\cdot\text{K}$$

$$\Delta G^\circ = 28150 \text{ J} \implies \Delta G^\circ > 0 \text{ nonspontaneous}$$

b) Determine the temperature in Celsius above which the reaction is spontaneous.

■ For spontaneous process

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ < 0$$

$$125000 - T \times 325 < 0 \implies T > \frac{125000}{325}$$

$$T > 385 \text{ K}$$

$$T > 112^\circ\text{C}$$



Sample Exercise

At 27°C, a reaction has $\Delta H = +10 \text{ kJ mol}^{-1}$ and $\Delta S = +30 \text{ J K}^{-1} \text{ mol}^{-1}$. What is the value of ΔG ?

$$\Delta G = \Delta H - T\Delta S \quad T = 300 \text{ K}$$

$$\Delta G = (+10 \text{ kJ mol}^{-1}) - (300 \text{ K})(+30 \text{ J K}^{-1} \text{ mol}^{-1})$$

$$\Delta G = +10 \text{ kJ mol}^{-1} - 9000 \text{ J mol}^{-1}$$

$$\Delta G = +10 \text{ kJ mol}^{-1} - 9 \text{ kJ mol}^{-1}$$

$$\Delta G = +1.0 \text{ kJ mol}^{-1}$$

Question

For the reaction

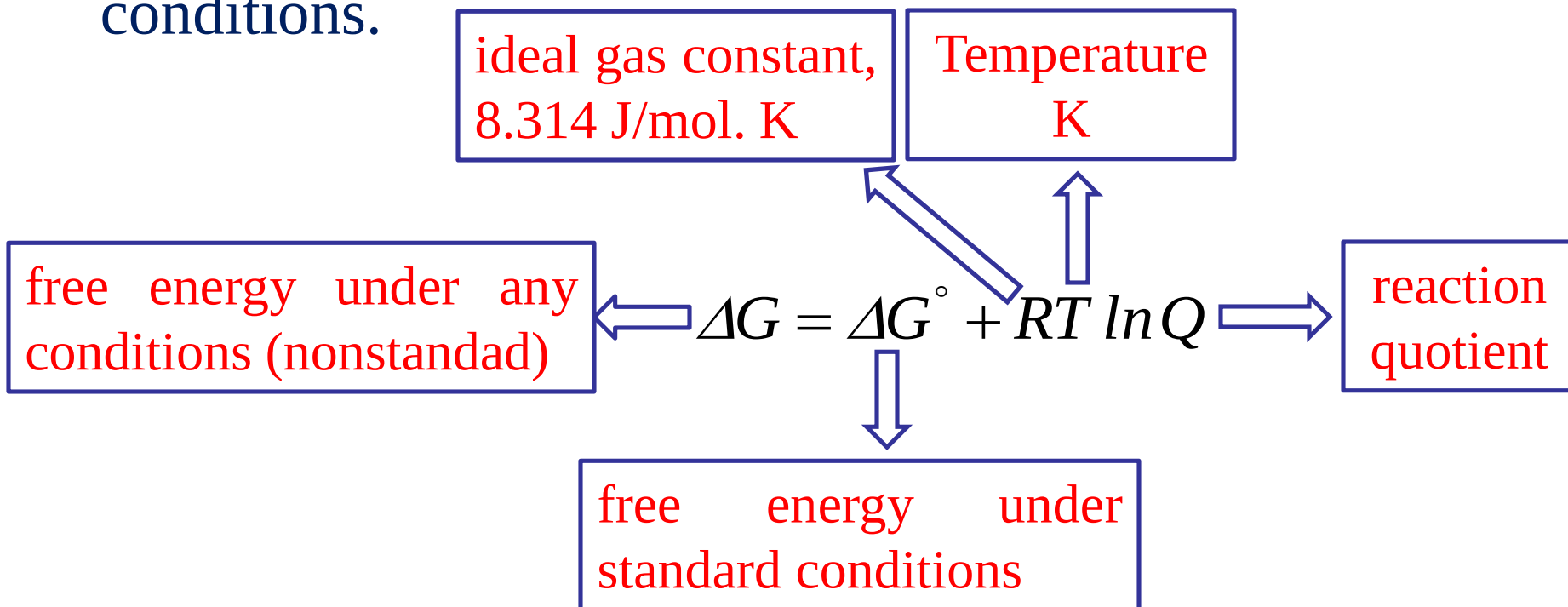


Calculate ΔG for the process at 298 K if $\Delta H = -144.3 \text{ kJ}$ and $\Delta S = -36.8 \text{ J K}^{-1}$

Answer: -133.3 kJ

19.7 Free Energy and the Equilibrium Constant

- Most chemical reactions occur under nonstandard conditions.



- For the reaction: $aA + bB \rightleftharpoons cC + dD$

$$Q = \frac{[C]^c [D]^d}{[A]^a [B]^b} \quad \text{concentration at any time} \quad K_{eq} = \frac{[C]_{eq}^c [D]_{eq}^d}{[A]_{eq}^a [B]_{eq}^b} \quad \text{all concentration at equilibrium}$$

19.7 Free Energy and the Equilibrium Constant

- Under standard conditions, the concentrations of all the reactants and products = 1. $\Rightarrow Q = 1 \Rightarrow \ln Q = 0$

$$\Delta G = \Delta G^\circ$$

- At equilibrium $\Rightarrow \Delta G = 0$ and $Q = K$

$$\Delta G = \Delta G^\circ + RT \ln Q$$

$$0 = \Delta G^\circ + RT \ln K$$

$$\Delta G^\circ = -RT \ln K \Rightarrow K = e^{-\Delta G^\circ / RT}$$

K	lnK	ΔG°	Result at equilibrium
> 1	+	-	Products are favored
$= 1$	0	0	neither products nor reactants are favored
< 1	-	+	Reactants are favored

Sample Exercise

- Given that ΔG_f° of HCl is 95.27 kJ/mol, calculate the equilibrium constant, K_p , for the following reaction at 25°C.



$$\begin{aligned}\Delta G^\circ_{\text{rxn}} &= \sum n \Delta G^\circ_{\text{products}} - \sum m \Delta G^\circ_{\text{reactants}} \\ &= \{1 \times 0 + 1 \times 0\} - \{2 \times 95.27\} \\ &= 190.54 \text{ kJ/mol}\end{aligned}$$

$$\Delta G^\circ = -RT \ln K$$

$$190.54 \text{ kJ/mol} = 8.314 \times 10^{-3} \text{ kJ/K}\cdot\text{mol} \times 298 \text{ K} \times \ln K_p$$

$$K_p = 3.98 \times 10^{-34} \quad K < 1 \text{ reactants are favored}$$

Sample Exercise

- Consider the reaction: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$,

A) If $\Delta G^\circ_{\text{rxn}} = -33 \text{ kJ}$, at 25°C , calculate the equilibrium constant, K .

$$\Delta G^\circ = -RT \ln K \longrightarrow \ln K^\circ = -\frac{\Delta G^\circ}{RT}$$

$$\ln K = -(-33 \times 10^3 \text{ J}) / (298 \text{ K} \times 8.314 \text{ J/K.mol}) = 13$$

$$K = e^{13} = 4.4 \times 10^5$$

(reaction goes to completion)

B) If $\Delta G^\circ_{\text{rxn}} = 12 \text{ kJ}$, at 250°C , calculate the equilibrium constant, K .

$$\Delta G^\circ = -RT \ln K \longrightarrow \ln K^\circ = -\frac{\Delta G^\circ}{RT}$$

$$\ln K = -(12 \times 10^3 \text{ J}) / (523 \text{ K} \times 8.314 \text{ J/K.mol}) = -2.8$$

$$K = e^{-2.8} = 0.061$$

(very little product is formed)